PENGUINS

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```
library(palmerpenguins)
library(ggplot2)
library(dplyr)
library(car)
library(tidyr)

# Load the dataset and remove rows with missing values
df_full <- penguins %>% drop_na()



## peek at total number of adorable penguins
table(df_full$species)
```

```
##
##    Adelie Chinstrap    Gentoo
##       146        68       119
```

Slide Demo

1) Calculatee Mean

```
## Randomly sample  68 penguins from each species (there are only 68 chinstraps)
set.seed(123) # random seed
df_sampled <- df_full %>%
  group_by(species) %>%
  sample_n(68) %>%
  ungroup()

## Preform Levene test
leveneTest(flipper_length_mm ~ species, data = df_sampled)
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value Pr(>F)
## group   2  0.3031 0.7388
##       201
```

```
## Determine mean flipper length for each species
df_sampled %>%
  group_by(species) %>%
  summarise(mean_flipper_length = mean(flipper_length_mm))
```

```
## # A tibble: 3 × 2
##   species   mean_flipper_length
##   <fct>                   <dbl>
## 1 Adelie                   190.
## 2 Chinstrap                196.
## 3 Gentoo                   217.
```

```
## Determine mean flipper length for all sampled penguins
df_sampled %>%
  summarise(mean_flipper_length = mean(flipper_length_mm))
```

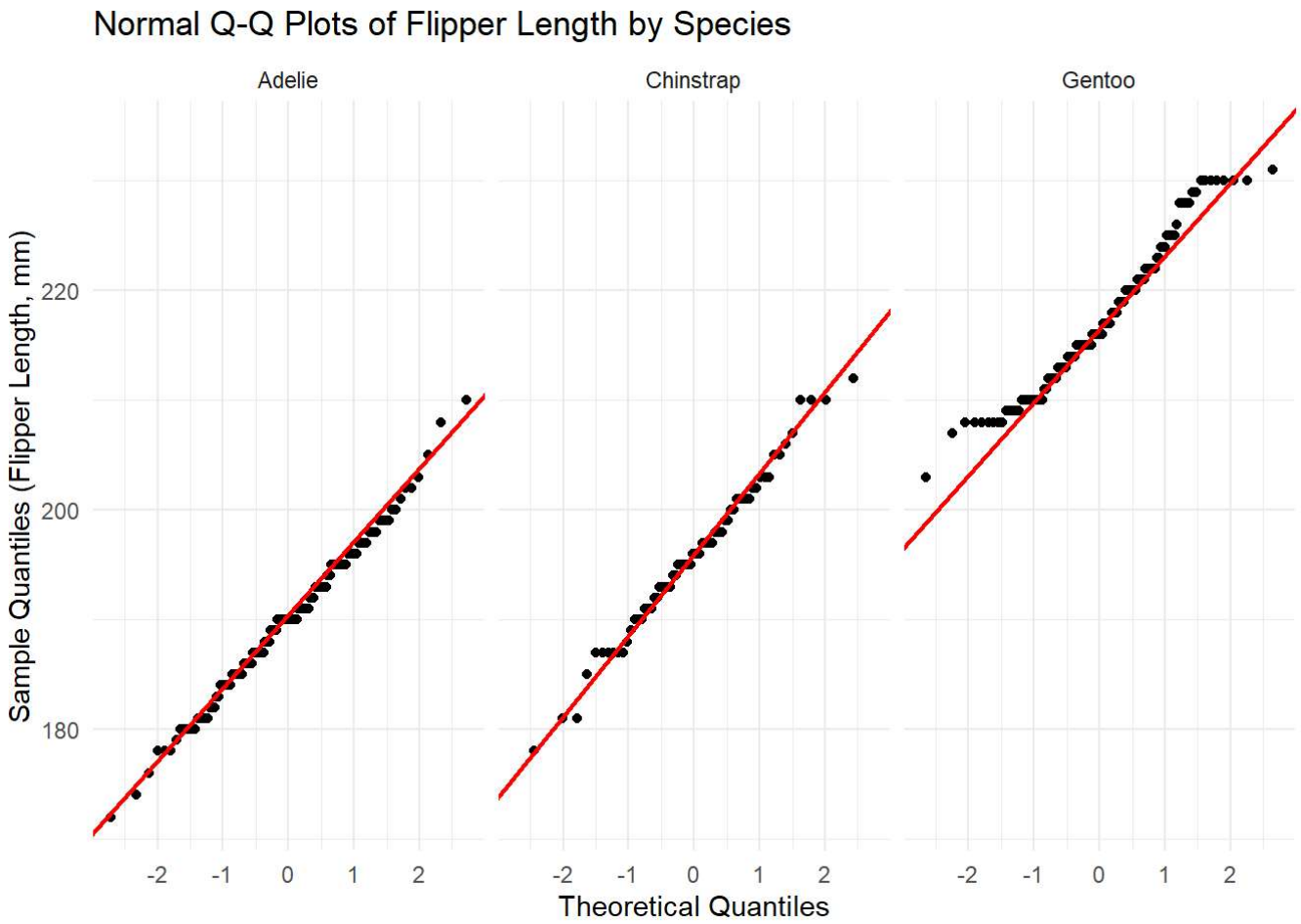
```
## # A tibble: 1 × 1
##   mean_flipper_length
##                 <dbl>
## 1                201.
```

1.1) Q-Q Plot

```
# Q-Q plots for eachspecies
ggplot(penguins, aes(sample = flipper_length_mm)) +
  geom_qq() +
  geom_qq_line(color = "red", linewidth = 0.8) +
  facet_wrap(~ species) +
  labs(
    title = "Normal Q-Q Plots of Flipper Length by Species",
    x = "Theoretical Quantiles",
    y = "Sample Quantiles (Flipper Length, mm)"
  ) +
  theme_minimal()
```

```
## Warning: Removed 2 rows containing non-finite outside the scale range
## (`stat_qq()`).
```

```
## Warning: Removed 2 rows containing non-finite outside the scale range
## (`stat_qq_line()`).
```



2) Calculate Sum of Squares Between Groups

```
grand_mean <- mean(df_sampled$flipper_length_mm)
ss_between <- df_sampled %>%
  group_by(species) %>%
  summarise(n = n(),
            mean_flipper_length = mean(flipper_length_mm)) %>%
  mutate(ss = n * (mean_flipper_length - grand_mean)^2) %>%
  summarise(ss_between = sum(ss))
ss_between
```

```
## # A tibble: 1 × 1
##   ss_between
##        <dbl>
## 1     27709.
```

3) Calculate Sum of Squares Within Groups

```
ss_within <- df_sampled %>%
  group_by(species) %>%
  summarise(n = n(),
            mean_flipper_length = mean(flipper_length_mm)) %>%
  left_join(df_sampled, by = "species") %>%
  mutate(ss = (flipper_length_mm - mean_flipper_length)^2) %>%
  group_by(species) %>%
  summarise(ss_within = sum(ss))
ss_within
```

```
## # A tibble: 3 × 2
##   species   ss_within
##   <fct>         <dbl>
## 1 Adelie        3285.
## 2 Chinstrap     3408.
## 3 Gentoo        2561.
```

```
## Total Sum of Squares Within Groups
ss_within_total <- sum(ss_within$ss_within)
ss_within_total
```

```
## [1] 9253.632
```

4) Variance (Mean Squares)

```
ms_between <- ss_between$ss_between / (length(unique(df_sampled$species)) - 1)
ms_within <- ss_within_total / (nrow(df_sampled) - length(unique(df_sampled$species)))
ms_between
```

```
## [1] 13854.31
```

```
ms_within
```

```
## [1] 46.03797
```

5) Calculate F-statistic

```
f_statistic <- ms_between / ms_within
f_statistic
```

```
## [1] 300.9322
```

6) Calculate F-statistic for ANOVA

```
anova_result <- aov(flipper_length_mm ~ species, data = df_sampled)
tab <- summary(anova_result)[[1]]

df1   <- tab[["Df"]][1]          # 2
df2   <- tab[["Df"]][2]          # 201
f_val <- tab[["F value"]][1]     # 300.933
p_val <- tab[["Pr(>F)"]][1]      # P-value from other variables and ANOVA table

alpha <- 0.05

# Formatted string for p-value presentation
p_str <- ifelse(p_val < 2.2e-16, "< 2.2e-16", sprintf("= %.3g", p_val))

# Print
sprintf("F(%d, %d) = %.1f, p %s", df1, df2, f_val, p_str)
```

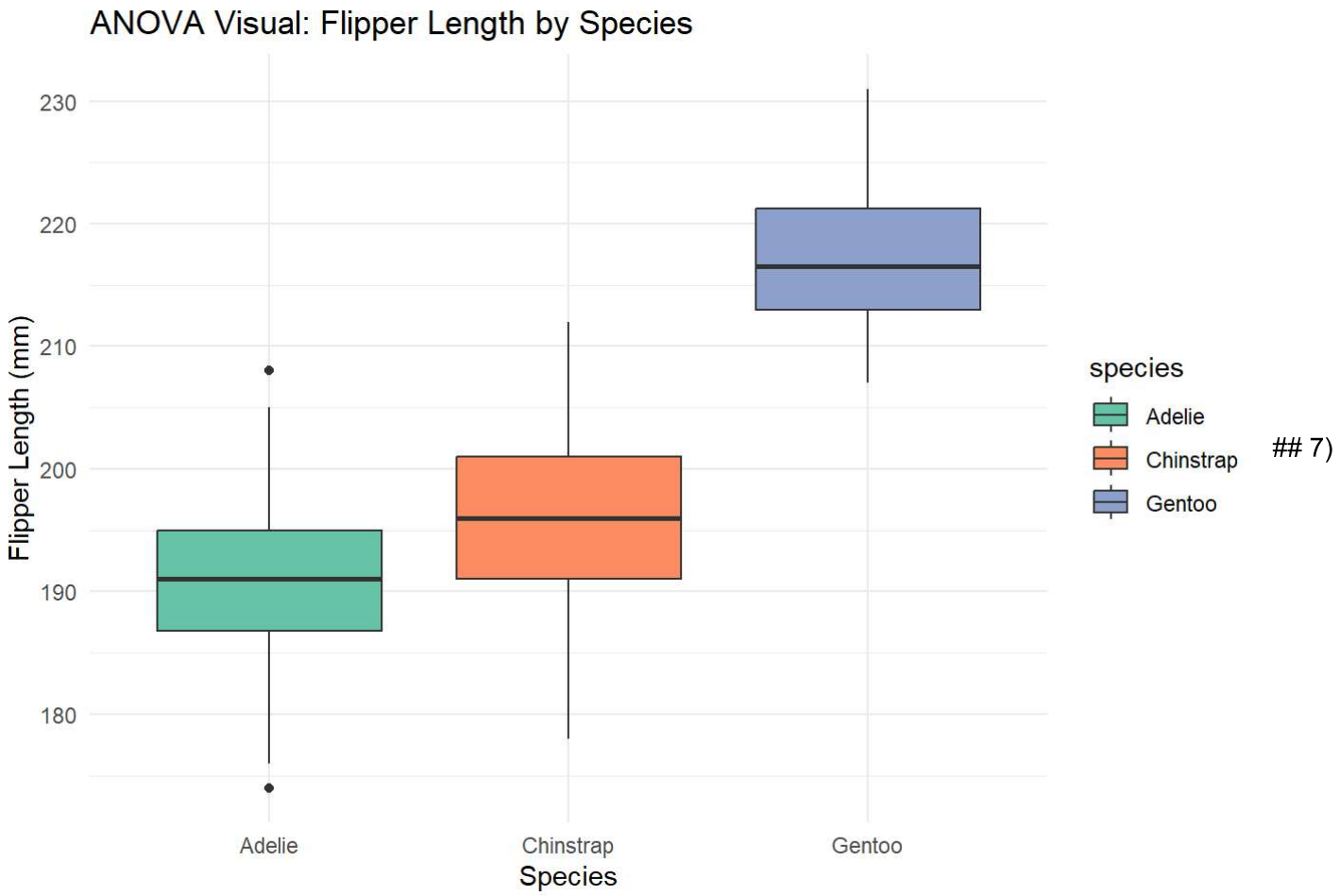
```
## [1] "F(2, 201) = 300.9, p < 2.2e-16"
```

```
# hypothesis check
if (p_val < alpha) {
  cat("There is a statistically significant difference in mean flipper length across specie
s.\n")
} else {
  cat("There is no statistically significant difference in mean flipper length across specie
s.\n")
}
```

```
## There is a statistically significant difference in mean flipper length across species.
```

Generate Visual

```
ggplot(df_sampled, aes(x = species, y = flipper_length_mm, fill = species)) +
  geom_boxplot() +
  labs(title = "ANOVA Visual: Flipper Length by Species",
       x = "Species",
       y = "Flipper Length (mm)") +
  theme_minimal() +
  scale_fill_brewer(palette = "Set2")
```



7)

Post-hoc Analysis

```
TukeyHSD(aov(flipper_length_mm ~ species, data = df_sampled))
```

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = flipper_length_mm ~ species, data = df_sampled)
##
## $species
##                       diff      lwr       upr    p adj
## Chinstrap-Adelie  5.338235  2.59064  8.085831 2.34e-05
## Gentoo-Adelie    26.955882 24.20829 29.703478 0.00e+00
## Gentoo-Chinstrap 21.617647 18.87005 24.365242 0.00e+00
```

Class Demo: In Built ANOVA Function

```
anova_result <- aov(flipper_length_mm ~ species, data = df_sampled) ## Variables flipper and spe
cies
summary(anova_result)
```

```
##              Df Sum Sq Mean Sq F value Pr(>F)
## species       2  27709   13854   300.9 <2e-16 ***
## Residuals   201   9254      46
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Generate Visual

```
ggplot(df_sampled, aes(x = species, y = flipper_length_mm, fill = species)) +
  geom_boxplot() +
  labs(title = "ANOVA Visual: Flipper Length by Species",
       x = "Species",
       y = "Flipper Length (mm)") +
  theme_minimal() +
  scale_fill_brewer(palette = "Set2")
```

